

CSE6624

Introduction to Web Programming (CityU)

Final Exam Project Brief · 40 points

Autumn semester 2026-2027 · Year 1 · Language of instruction: English

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Allowed stack: HTML5 · CSS (Flexbox/Grid) · Bootstrap (optional) · vanilla JavaScript · DOM · localStorage

Forbidden: React, Vue, Angular, Node.js, Express, databases, WordPress, site builders

This PDF = Final Exam only (40 pts)

Attestations, SIS and TSIS live in a separate booklet. Practice/Lab weekly scores are separate.

Chill tone

This is a guided project, not a trick exam. Start early, keep screenshots, use your personal seed. AI helpers are OK if you can explain and slightly modify your code in the oral defence.

Contents: 1 Goals · 2 Scope & forbidden tech · 3 Timeline · 4 Themes & seed · 5 Must-have features · 6 Rubric (40) · 7 Defence guide · 8 Example walkthrough · 9 Submission · 10 FAQ · 11 Beginner steps · 12 Self-score · 13 Integrity

Quick map of this brief

#	Section	One-liner
1	Goals	What success looks like
2	Scope & forbidden tech	HTML/CSS/JS only
3	Timeline	Day-by-day beginner plan
4	Themes & seed	24 niches + ID formula
5	Must-have features	HTML, CSS, forms, JS, a11y
6	Rubric	40 pts bands
7	Defence	Demo script + AI policy
8	Example walkthrough	Film club sample
9	Submission checklist	README + screens
10	Late note & FAQ	Soft policy + contact
11	Beginner steps	First-weekend order
12	Self-score sheet	Quality bar
13	Peer integrity	Anti-copy reminders

Attestations / SIS / TSIS: see the separate student pack. This brief is exam-only.

1. Goals of the exam project

By the end of this project you prove that you can build a small, real-looking multi-page website for one niche, with clean structure, responsive layout, forms, and JavaScript interactivity. You also prove ownership: personal seed, unique niche, screenshots, GitHub README with your name + ID, and a short oral defence.

- **Learning goal:** semantic HTML + CSS layout + vanilla JS DOM skills from the semester.
- **Product goal:** a niche site a visitor could actually browse (Home, About/Services, Gallery or List, Contact).
- **Integrity goal:** your work is personally watermarked and defensible live.
- **Non-goal:** a production backend, React SPA, or paid hosting (GitHub Pages or local demo is enough).

RU tip / Цель

Экзамен = один законченный мини-сайт на HTML/CSS/JS. Не React и не Node. Главное: структура, адаптив, форма, JS-интерактив и защита 5-7 минут.

2. Scope and forbidden technology

2.1 What you must build

A multi-page (or multi-section SPA-like) static website for **one** niche from the list (or an approved unique free-choice niche). At least **4 distinct pages or major sections** with shared navigation. Content must be niche-specific (not lorem-only).

2.2 Allowed tools

- HTML5, CSS3 (custom properties, Flexbox, Grid), optional Bootstrap 5 CDN or local files.
- Vanilla JavaScript (ES6+ fine): DOM, events, forms, localStorage, simple arrays/objects.
- VS Code / Cursor, Git + GitHub, Chrome/Firefox DevTools, Live Server or similar.
- Free image assets with attribution; your own photos/screenshots preferred.

2.3 Forbidden (zero or heavy deduction)

- React, Vue, Angular, Svelte, Next.js, or any component framework.
- Node.js / Express / any server-side runtime as part of the exam deliverable.
- SQL / Mongo / Firebase / Supabase as a required backend (static JSON file or localStorage is OK).
- WordPress, Wix, Tilda, Webflow, Framer, AI site builders that spit full sites for you.
- Copying a classmate repo and only renaming titles / niche strings.
- Submitting only Attestation 1/2 work without exam-level upgrades (exam must be a clear final product).

Warning

If Temirlan opens your zip and sees create-react-app or package.json with react, the stack part of the rubric collapses. Stay on HTML/CSS/JS.

3. Suggested timeline (exam period)

Exact calendar dates follow the university exam schedule. Use this as a beginner-friendly plan once the exam window opens (often overlapping late Week 15 / exam week).

When	Focus	Done looks like
Day 1	Seed + niche claim	Compute Hue/Accent; write README stub; create exam branch/folder
Day 2-3	HTML skeleton	4+ pages, nav, footer watermark, semantic tags
Day 4-5	CSS / responsive	Mobile-first; seed colors; Flex/Grid; optional Bootstrap
Day 6-7	Forms + JS	Validate form; interactive list/filter/toggle; localStorage optional
Day 8	A11y + polish	Labels, alt text, focus, contrast; remove broken links
Day 9	Docs + screens	README full; VS Code + DevTools screenshots; zip
Defence day	Demo 5-7 min	Run live; answer modify questions

RU tip / План

Не начинайте с красоты. Сначала HTML + watermark + README, потом CSS, потом JS. Скриншоты делайте по ходу, не в последнюю ночь.

4. Themes, seed, and how to pick

Anti-copy personalization is mandatory for the exam (same idea as attestations, exam-level finish).

4.1 Personal seed formula

ID2 = last 2 digits of student ID (00-99). **M** = birth month (1-12). **L** = first letter of surname (A=1 ... Z=26; Latin transliteration for non-Latin names).

1. **Hue** = $(ID2 \times 7 + M \times 13) \bmod 360 \rightarrow$ primary HSL hue.

2. **Accent** = $(Hue + 40 + L) \bmod 360$.

3. **Niche index** = $(ID2 + M + L) \bmod 24 \rightarrow$ niche number = index + 1 from the table below.

4. If that niche is already taken in your group, take the **next free** niche and write both numbers in README.

5. **Watermark (exact)**: CSE6624 | FULL_NAME | STUDENT_ID | niche-N

6. Show watermark in page footer AND in README. Seed colors must drive CSS variables.

4.2 Free choice uniqueness

You may invent your own niche instead of the list **only if** (a) it is Almaty/campus-flavored or clearly local, (b) nobody else in the group has the same idea, (c) you email Temirlan a one-line claim before defence week, (d) README explains why it is unique. Copying a famous global brand homepage is not unique.

4.3 Niche list (20+ options)

Niches A	Niches B
1. Specialty coffee cafe near Kok-Tobe	13. Mental-health peer support club
2. Student fitness / gym club (Satbayev)	14. Campus book swap network
3. Pet shelter & adoption (Almaty)	15. Electric scooter rental (city)
4. University esports team hub	16. Traditional crafts marketplace
5. Hiking & weekend travel (Alatau)	17. Campus food-waste reduction
6. Campus language exchange club	18. Amateur astronomy club nights
7. Board-game cafe (Almaty centre)	19. Volunteer blood-donor drive
8. Sustainable fashion thrift shop	20. Mobile repair workshop
9. Street-food map (Green Bazaar area)	21. Plant nursery / balcony garden
10. Coding / robotics school for kids	22. Film club & short-movie nights
11. Indie music venue calendar	23. Chess club with ladder board
12. Photography portfolio studio	24. Homemade bakery brand site

4.4 Worked seed example

Fake student: **Aigerim Saduakas**, ID ending **47**, born March (**M=3**), surname **S** → **19**.

1. Hue = $(47 \times 7 + 3 \times 13) \bmod 360 = 368 \bmod 360 = 8$

2. Accent = $(8 + 40 + 19) \bmod 360 = 67$

3. Niche index = $(47 + 3 + 19) \bmod 24 = 69 \bmod 24 = 21 \rightarrow$ niche **22. Film club & short-movie nights**

4. CSS: --primary: hsl(8,65%,40%); --accent: hsl(67,70%,45%);

RU tip / Seed

Посчитайте один раз, запишите в README. Цвета primary/accent обязательны. Watermark должен читаться на скриншотах VS Code и браузера.

5. Must-have features checklist

Use this as your build checklist. Missing a major item costs rubric points.

5.1 Semantic HTML

- ☐ Meaningful `<header>` `<nav>` `<main>` `<section>` `<article>` `<footer>`
- ☐ One `<h1>` per page; heading levels not skipped randomly
- ☐ Lists for menus / features; tables only when data is tabular
- ☐ Images have useful alt; decorative images use empty alt
- ☐ Shared nav links work between all pages

5.2 Responsive CSS

- ☐ Viewport meta tag present
- ☐ Looks usable at ~375px and ~1200px widths (test in DevTools)
- ☐ Flexbox and/or Grid used intentionally (not only float hacks)
- ☐ Seed CSS variables for primary/accent used in buttons, links, headings
- ☐ No horizontal scrollbar on mobile for main content
- ☐ Bootstrap optional: if used, customize theme colors toward your seed (not default purple-blue clone)

5.3 Forms

- ☐ At least one real form (contact, booking, signup, feedback)
- ☐ Labels associated with inputs; required fields marked
- ☐ Client-side validation in JS (empty, email pattern, or length) with clear messages
- ☐ Success state after valid submit (message or UI change; no real email server required)

5.4 JavaScript interactivity

- ☐ At least **two** meaningful interactions (examples: filter list, add/remove cards, modal, accordion, theme toggle, quiz, cart-like counter)
- ☐ No inline `onClick=...` soup for everything; prefer `addEventListener`
- ☐ JS fails gracefully if an element is missing (no blank white page from one typo)
- ☐ Optional: `localStorage` to remember a preference or list

5.5 Accessibility (a11y) basics

- ☐ Keyboard: can tab to links, buttons, form fields
- ☐ Visible focus outline not removed without replacement
- ☐ Color contrast roughly readable (avoid pale gray on white)
- ☐ Form errors announced in text, not only by color
- ☐ Language attribute on `<html lang="en">` (or appropriate)

5.6 Recommended page set

Page / section	Minimum content
Home	Hero, short pitch, CTA, 3 feature cards
About / Services	Story of the niche + offerings
Gallery / Catalog / Events	Grid of items; filter or search with JS
Contact / Join	Form + map placeholder or address + socials
(Optional) FAQ or Blog	Accordion or article list

RU tip / Фичи

Минимум: 4 раздела, форма с валидацией, 2 JS-фичи, адаптив, watermark, README. Bootstrap не обязателен; чистый CSS тоже ок.

5b. Content writing and a11y deep dive

Empty lorem sites lose Content/UX points. Write short, specific copy for your niche.

Content mini-rules

- Replace 'Welcome to our website' with a niche promise (who + what + where in Almaty/campus).
- Three feature cards should answer: why join, what happens, when/where.
- Contact page needs a human path: hours, place, Telegram/email placeholder.
- If you use stock photos, one line of credit in README is enough.

A11y quick test (10 minutes)

1. Unplug mouse: tab through nav and form. Can you see focus?
2. Zoom to 150%: does layout still work?
3. Turn on grayscale briefly: are errors only red without text?
4. Read one page with headings outline (DevTools or extension): is there a single h1?
5. Check one image alt: does it describe meaning, not 'image1'?

RU tip / Контент

Лучше 120 своих слов про клуб, чем 800 слов lorem. Адаптив и a11y проверяйте в DevTools.

5c. JS pattern menu (pick two)

Pattern	Technique	Fits
Filter / search cards	data-* attributes + hidden toggle	Catalog, events, pets
Add / remove list	array in memory + render()	Menu, tasks, members
Modal / lightbox	dialog or div + Escape to close	Gallery posters
Accordion FAQ	aria-expanded toggle	About / Join
Theme toggle	class on body + localStorage	Any niche
Simple quiz	score state + result panel	Clubs / education niches
Form wizard 2 steps	show/hide fieldsets	Booking / join
Sort by date/price	array.sort + re-render	List pages

Two solid patterns beat five half-broken ones. Form validation counts as part of JS credit but you still need a second interactive feature beyond the form.

6. Rubric (Total = 40 points)

Final exam grade for CSE6624 is this project + oral defence. Bands below guide partial credit.

Criterion	Pts	What we look for
A. Structure & semantic HTML	8	Pages, nav, semantics, valid-ish markup, no broken IA
B. CSS & responsive	8	Layout, seed colors, mobile/desktop, polish
C. JavaScript interactivity	8	Two+ features, form validation, clean events
D. Content & UX	6	Niche-specific text/images, clarity, a11y basics
E. GitHub & documentation	4	README name+ID+seed+run; folder; screenshots
F. Oral defence	6	5-7 min demo + explain/modify answers
TOTAL	40	

6.1 Band guide (examples)

Area	Strong	Mid	Weak
A Structure (8)	7-8: clear multi-page IA, semantic tags, consistent nav	4-6: works but messy headings/div soup	0-3: single broken page / missing sections
B CSS (8)	7-8: responsive + seed theme + intentional layout	4-6: desktop OK, weak mobile or default Bootstrap only	0-3: unstyled or broken
C JS (8)	7-8: two solid features + form validation	4-6: one feature or validation half-done	0-3: no JS or copy-paste dead code
D Content (6)	5-6: niche voice, useful copy, a11y care	3-4: generic lorem with few edits	0-2: empty / irrelevant
E Docs (4)	4: README complete + screenshots + watermark	2-3: README thin or missing screens	0-1: no README / no identity
F Defence (6)	5-6: confident demo + can modify	3-4: demo OK, shaky explanations	0-2: cannot explain own code

Integrity deductions (can stack): identical peer structure (–up to 10), missing seed/watermark (–3), forbidden stack (–up to 15), no defence attendance without excuse (defence = 0).

OK check

8+8+8+6+4+6 = 40. Keep this split when you self-grade before submit.

7. How to defend (5-7 minutes)

7.1 Demo script (practice aloud)

- 1. 0:00-0:30** State your name, ID, niche number, and seed hues (show README).
- 2. 0:30-1:30** Open Home: explain audience and CTA; show watermark in footer.
- 3. 1:30-3:00** Walk two key pages (catalog/events + contact). Show responsive resize once.
- 4. 3:00-4:30** Live JS: trigger filter / add item / toggle; submit the form with a bad then good value.
- 5. 4:30-5:30** Open DevTools briefly (or screenshot folder) proving it is your work.
- 6. 5:30-7:00** Answer Temirlan's questions; be ready to edit one small thing live.

7.2 What Temirlan typically asks

- Why did you choose this niche / what is unique vs a classmate?
- Where is your seed used in CSS? Change primary button color live.
- Explain one JS function: what parameters, what DOM nodes?
- How does your form validation work? Break it on purpose and fix.
- Which semantic tags did you use and why not only div?
- What would you add next week if this were a real client site?

7.3 Common fails

- Demo only works on the student's laptop with mystery local paths; relative links broken in zip.
- Cannot find own JS file during defence.
- README has no name/ID; watermark missing on screenshots.
- Everything is Bootstrap demo pages with niche title swapped.
- AI-generated wall of code the student cannot trim or rename.
- Claiming 'the form sends email' when there is no backend (say: client-side success message).

7.4 AI policy (exam)

ChatGPT, Copilot, and similar tools are **allowed as helpers**. You must understand every line you submit. Explain + modify on request is the bar. Blind paste of a full tutorial site with renamed headings is not enough. If AI wrote a function, you should still be able to change a selector, rename a variable, or fix a bug live.

RU tip / Защита

Репетируйте вслух 2 раза. Держите открытыми: сайт, README, один JS-файл. Если волнуетесь - нормально; Temirlan не устраивает унижение, но вопросы будут.

8. Example project walkthrough (sample niche)

Sample niche for this walkthrough: **#22 Film club & short-movie nights** (from the worked seed: Aigerim Saduakas). Your niche will differ; copy the process, not the content.

8.1 Folder sketch

```
cse6624-exam-aigerim/
index.html          # Home
about.html          # Club story + team
screenings.html     # Upcoming nights + JS filter
join.html           # Contact / membership form
css/styles.css       # seed variables + layout
js/main.js          # nav highlight, filter, form
assets/             # posters (credited)
screenshots/        # vscode.png, form-valid.png, mobile.png
README.md           # name, ID, seed, how to open
```

8.2 Wireframe checklist (text wireframes)

Home (index.html)

- [Nav] Home | About | Screenings | Join ... watermark footer always
- [Hero] Title: Alatau Short Nights · CTA: See this month's films
- [3 cards] Community · Student directors · Free campus hall
- [Teaser list] Next 3 screenings (static HTML, linked to Screenings)

Screenings (screenings.html)

- [Filter bar] All | Comedy | Drama | Doc (JS shows/hides cards by data-genre)
- [Card grid] poster, title, date, hall, 1-line blurb
- [Empty state] If filter matches nothing, show a friendly message

Join (join.html)

- [Form] name, email, year of study, favorite genre, message
- [JS] block submit if email invalid; show green success box on OK
- [Aside] club Telegram / meeting room (fake but consistent)

8.3 Fake screenshot list (include similar)

File	What it proves
01-vscode-home.png	VS Code with index.html + watermark comment or footer visible
02-devtools-mobile.png	Chrome device mode ~375px on Home
03-filter-js.png	Screenings filtered to Drama
04-form-error.png	Invalid email message visible
05-form-ok.png	Success state after valid submit
06-readme.png	GitHub README with name, ID, niche-22, hues 8/67

8.4 Mini JS sketch (illustrative)

```
// Filter screening cards by data-genre
document.querySelectorAll('[data-filter]').forEach(btn => {
  btn.addEventListener('click', () => {
    const g = btn.dataset.filter;
    document.querySelectorAll('.card').forEach(card => {
      const show = (g === 'all') || card.dataset.genre === g;
      card.hidden = !show;
    });
  });
});
```

RU tip / Пример

Это образец процесса. Не сдавайте тот же Film club, если ваш seed указывает другой niche (или докажете free-choice uniqueness).

9. Submission checklist

Submit as announced (usually GitHub link + zip + LMS if required). Before you upload:

- ☐ Repo or zip opens with relative paths (no C:\Users\... links)
- ☐ README.md includes: full name, student ID, group, niche number + title, Hue/Accent, how to open (Live Server / double-click), AI tools used (if any)
- ☐ Watermark string on every page footer
- ☐ screenshots/ folder with at least 4 images (VS Code, mobile, JS feature, form)
- ☐ 4+ pages/sections with working nav
- ☐ Form validation works offline
- ☐ No React/Node; no secrets; no huge unused node_modules
- ☐ You can run the demo in under 30 seconds on defence day

9.1 README template (copy and fill)

```
# CSE6624 Exam Project
Student: FULL_NAME
ID: STUDENT_ID
Group: ...
Niche: N - Title
Seed: Hue=..., Accent=..
Watermark: CSE6624 | FULL_NAME | STUDENT_ID | niche-N

## How to run
Open index.html via Live Server (or any static server).

## Features
- ...
## AI tools used (if any)
- ... (what you asked help for)
## Screenshots
See /screenshots
```

10. Soft late note, FAQ, contacts

10.1 Late policy (soft reminder)

Aim for the announced exam deadline. Work late without a documented excuse typically loses **10%** of the exam item. More than about one week late may be zero for the exam (instructor decision, syllabus spirit). If you are sick or have an official conflict, write early.

10.2 FAQ

- **Can I reuse Attestation 1/2?** Yes as a starting base, but the exam must clearly grow (more pages, stronger JS, better content, full README + defence readiness).
- **Bootstrap required?** No. Optional.
- **GitHub private OK?** Yes if you add the instructor; public is fine too.
- **Team project?** Individual unless Temirlan explicitly allows pairs for your group.
- **Kazakh/Russian UI?** English is the course language; short bilingual labels are fine if consistent.
- **Where are SIS/Attestation rubrics?** Separate PDF: CSE6624_Attestations_and_SIS_TSYS.pdf.

10.3 Contact

Questions: **shoinbektima@gmail.com** (subject line: CSE6624 Exam + your group). Bring curiosity, not panic. See you at the defence.

You have got this

Small steps beat all-nighters. Seed → HTML → CSS → JS → screenshots → rehearsal. 40 points are earned by clear ownership and a working niche site.

11. Beginner step-by-step (first weekend)

If this is your first multi-page site, follow this order. Do not skip README and watermark.

1. Create folder `cse6624-exam-YOURNAME` and `git init`.
2. Write README stub with name, ID, niche, seed numbers (even if CSS is empty).
3. Create `index.html` with header/nav/main/footer and the watermark string in the footer.
4. Duplicate the skeleton into about / catalog / contact pages; fix nav links.
5. Add `css/styles.css` with `:root { --primary; --accent; }` from your Hue/Accent.
6. Style a simple hero + card grid with Flexbox or Grid; test 375px width.
7. Add contact form HTML with labels; wire JS validation in `js/main.js`.
8. Add second JS feature (filter or add-to-list). Commit after each working feature.
9. Take screenshots; paste links into README; zip and upload as announced.

RU tip / Новичкам

Один коммит = одна маленькая победа. Если застряли на CSS - сначала работающий HTML и форма.

12. Quality bar and self-score sheet

Before defence, score yourself honestly. Aim for 32+ before polish anxiety.

Band	Self-check statement	My pts
A Structure /8	I have 4+ sections, semantic tags, working nav	
B CSS /8	Seed colors + responsive + no huge overflow	
C JS /8	Two features + form validation demoable	
D Content /6	Niche text (not pure lorem) + a11y basics	
E Docs /4	README complete + 4 screenshots + watermark	
F Defence /6	I rehearsed the 5-7 min script twice	
Sum /40		

12.1 Polish checklist (last 90 minutes)

- ☐ Spell-check niche title and your name in README
- ☐ Remove console.log spam and commented-out junk
- ☐ Fix any 404 image or CSS path
- ☐ Confirm watermark on every page
- ☐ Open zip on a second folder to prove relative paths work
- ☐ Write three answers: seed math, one JS function, one CSS layout choice

13. Peer difference and integrity reminders

Temirlan may spot-check two repos side by side. Make your project look like **yours**.

- Different niche content blocks (not the same three feature sentences).
- Different component/class naming (avoid identical hero-blob-22 copies).
- Different screenshot set showing your watermark and ID.
- If you pair-programmed for learning, still submit separate niches and separate codebases.
- Cite image sources in README if not your photos.

Warning

Sharing a finished zip in the group chat so friends can 'just change the title' hurts everyone. Help peers with concepts; do not gift full projects.

13.2 What 'good enough' looks like

A calm, complete niche site with clear ownership beats a half-broken 'animation festival'. Prefer readable CSS and two solid JS features over five broken libraries.